Intersection of a Line and a Plane

Knowledge/ Understanding

1. For each of the following, find the intersection of the line and the plane.
   a. $x = 4 - t, y = 6 + 2t, z = -2 + t$ and $2x - y + 6z + 10 = 0$
   b. $x = 3 + 4t, y = -2 - 6t, z = \frac{1}{2} - 3t$ and $3x + 4y - 7z + 7 = 0$
   c. $x = 5 + t, y = 4 + 2t, z = 7 + 2t$ and $2x + 3y - 4z + 7 = 0$
   d. $\vec{r} = (2, 14, 1) + t(-1, -1, 1)$ and $3x - y + 2z + 6 = 0$
   e. $\vec{r} = (5, 7, 3) + t(0, 1, -1)$ and $z + 5 = 0$

2. a. Does the line $\vec{r} = (-2, 6, 5) + t(3, 2, -1)$ lie in the plane $3x - 4y + z + 25 = 0$?
   b. Does the line $\vec{r} = (4, -1, 2) + t(3, 2, -1)$ lie in the plane $3x - 4y + z - 17 = 0$?

3. Where does the plane $3x - 2y - 7z - 6 = 0$ intersect
   a. the x-axis? b. the y-axis? c. the z-axis?

Part B

4. a. In what point does the plane $\vec{r} = (6, -4, 3) + s(-2, 4, 7) + t(-7, 6, -3)$ intersect
   i) the x-axis ii) the y-axis iii) the z-axis

5. Where does the line $\vec{r} = (6, 10, 1) + t(3, 4, -1)$ meet
   a. the xy-plane b. the xz-plane c. the yz-plane

Communication

6. State whether it is possible for the lines and planes described below to intersect in one point, in an infinite number of points, or in no points.
   a. a line parallel to the x-axis and a plane perpendicular to the x-axis
   b. a line parallel to the y-axis and a plane parallel to the y-axis
   c. a line perpendicular to the z-axis and a plane parallel to the z-axis

Application

7. Find the point of intersection of the plane $3x - 2y + 7z - 31 = 0$ with the line that passes through the origin and is perpendicular to the plane.

Application

8. Find the point at which the normal to the plane $4x - 2y + 5z + 18 = 0$ through the point $(6, -2, -2)$ intersects the plane.

11. For what values of k will the line $\frac{x-k}{3} = \frac{y+4}{2} = \frac{z+6}{1}$ intersect the plane $x - 4y + 5z + 5 = 0$

    a. in a single point?

    b. in an infinite number of points?

    c. in no points?

Answers

1. a. $(4, 6, -2)$ **b.** $(1, 1, 2)$ **c.** no intersection
d. $(x, y, z) = (2 - t, 14 - t, 1 + t)$ **e.** $(5, 15, -5)$
2. a. yes **b.** no **3. a.** $(2, 0, 0)$ **b.** $(0, -3, 0)$ **c.** $\left(0, 0, \frac{-6}{7}\right)$
4. a. **(i)** $\left(\frac{28}{27}, 0, 0\right)$ **(ii)** $\left(0, \frac{56}{55}, 0\right)$ **(iii)** $\left(0, 0, \frac{-7}{2}\right)$
b. **(i)** $\vec{r} = (-1, 2, 0) + k(-55, 54, 0)$
(ii) $\vec{r} = (0, 8, 24) + p(0, 16, 55)$
(iii) $\vec{r} = (4, 0, 10) + u(8, 0, 27)$ **5. a.** $(9, 14, 0)$
b. $\left(\frac{-3}{2}, 0, \frac{7}{2}\right)$ **c.** $(0, 2, 3)$ **6. a.** one point **b.** no points, or an infinite number of points
c. no points, one point, or an infinite number of points
7. $\left(\frac{3}{2}, -1, \frac{7}{2}\right)$ **8.** $\left(\frac{14}{5}, \frac{-2}{5}, -6\right)$ **9. a.** 1, 4, 3 **b.** $5, \frac{-5}{2}, -5$
11. a. no value **b.** $k = 9$ **c.** $k \neq 9$